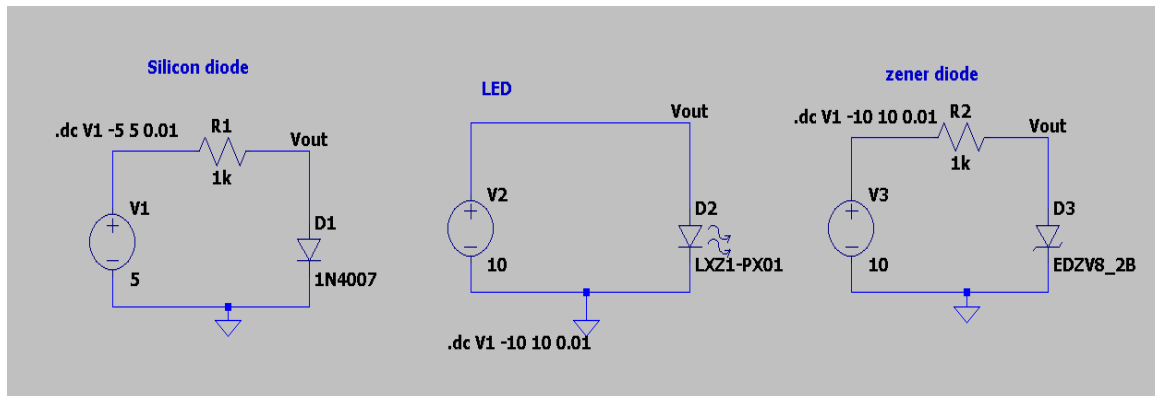


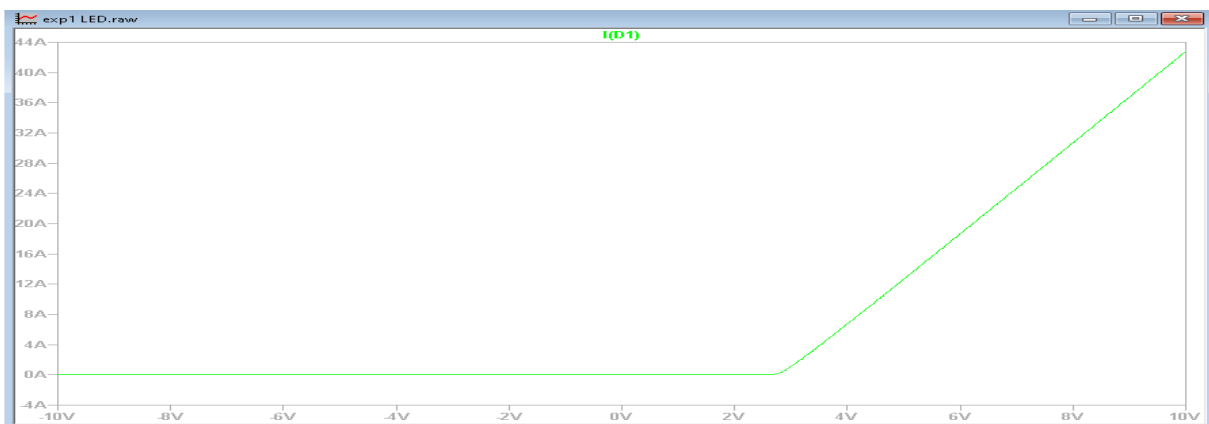
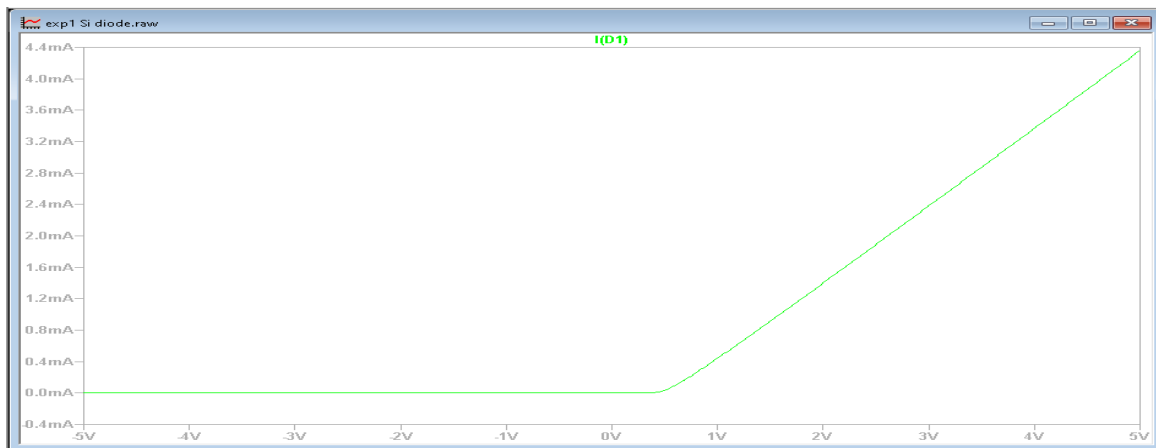
LAB FILE - ELECTRONIC DEVICES & CIRCUITS

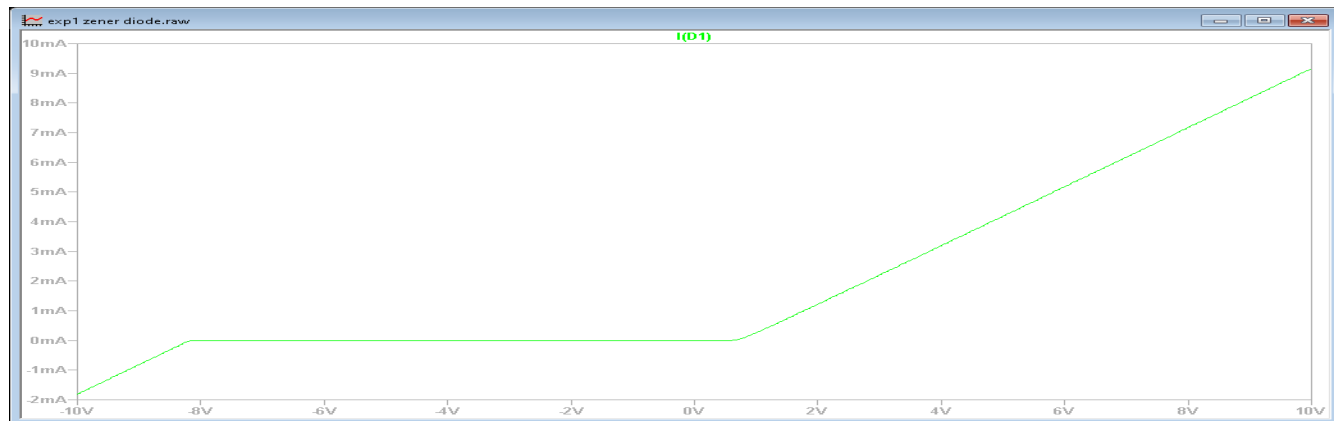
Experiment – 1

- **Aim :** Plot the input and output characteristics of a silicon diode, LED and zener diode.
- **Software used :** Lt-spice
- **Theory :**
 - **Silicon diode :** A silicon diode is a semiconductor device that allows current to flow in only one direction. It consists of a PN-junction made from silicon material. When forward biased, it conducts electricity with a voltage drop of approximately 0.7V. In reverse bias, it blocks current flow until the reverse breakdown voltage is reached. Silicon diodes are widely used in rectifiers, clippers, and voltage regulation circuits due to their reliability and efficiency.
 - **LED :** An LED is a special type of diode that emits light when forward biased. It is made from materials like gallium arsenide (GaAs) and gallium phosphide (GaP), which determine the color of the emitted light. The forward voltage drop of an LED ranges from 1.8V to 3.3V, depending on the color. LEDs work on the principle of electron-hole recombination, where energy is released in the form of photons (light). They are commonly used in displays, indicator lights, and lighting systems due to their high efficiency and low power consumption.
 - **Zener Diode :** A Zener diode is designed to operate in reverse bias mode without damage. Unlike a regular diode, it allows current to flow in the reverse direction once the applied voltage exceeds the Zener breakdown voltage (typically between 2V and 200V, depending on the diode). This unique property makes it ideal for voltage regulation and overvoltage protection in circuits. In forward bias, it behaves like a normal silicon diode with a voltage drop of around 0.7V.
- **Implemented Schematic :**



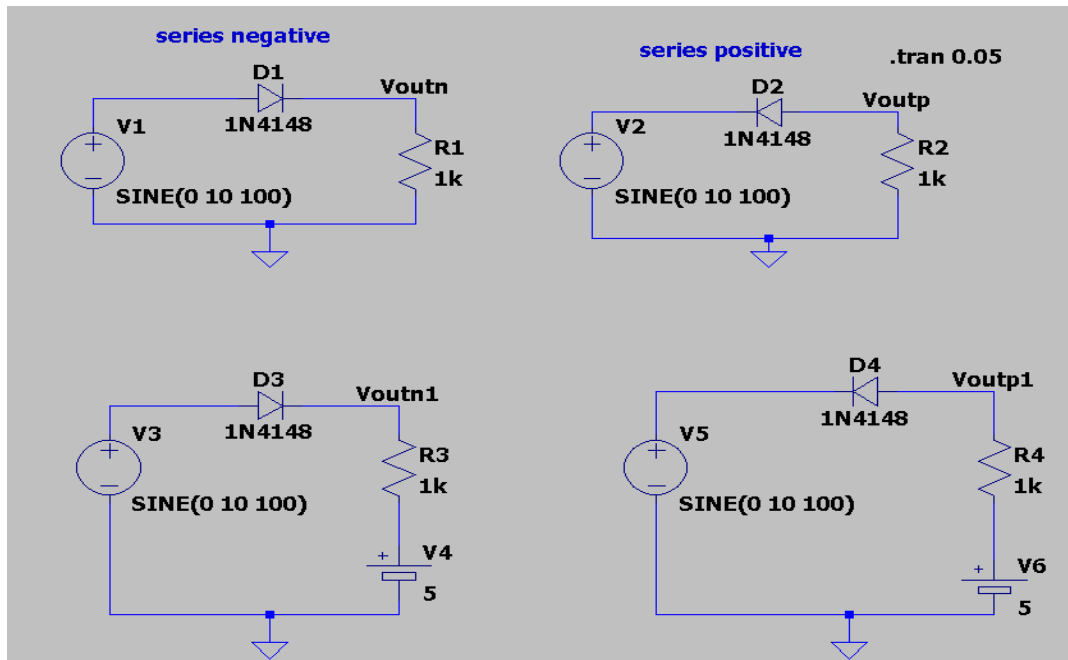
- **Obtained Waveforms :**



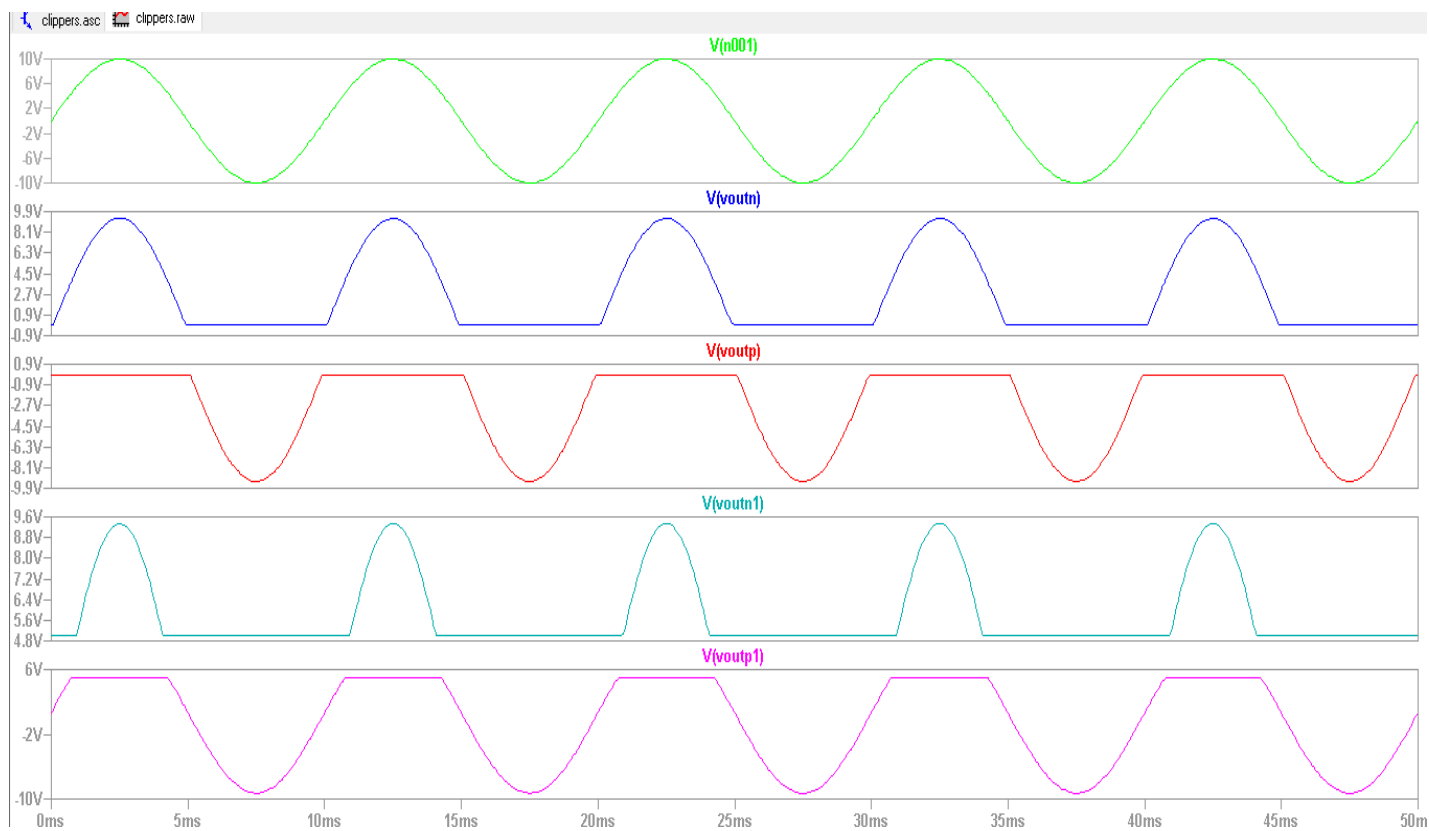


EXPERIMENT – 2

- **Aim :** To implement series and parallel clipper circuits using diodes and resistors.
- **Software used :** Lt-spice
- **Theory :** A clipper circuit is a nonlinear circuit that removes or "clips" a portion of an input signal without distorting the remaining waveform. These circuits are widely used in signal shaping, waveform limiting, and voltage protection applications. Clippers are classified into series clippers and parallel clippers, based on how the diode is connected in the circuit.
 - **Series Clipper :** In a series clipper, the diode is connected in series with the load. Depending on the diode's orientation, the circuit can clip either the positive or negative half of the input signal. When the diode is forward biased, it conducts, allowing the input signal to pass through. In reverse bias, it blocks the current, thereby clipping the unwanted portion of the signal. Series clippers are commonly used in signal processing circuits to limit voltage swings.
 - **Parallel Clipper :** In a parallel clipper, the diode is placed in parallel with the load. Here, the diode either shunts the positive or negative part of the waveform, preventing it from reaching the output. A resistor is used in series with the input to control current flow. Similar to series clippers, the diode's orientation determines whether the positive or negative portion of the waveform is clipped. Parallel clippers are often used for signal shaping and protection circuits.
 - Both series and parallel clippers can be modified using biasing voltage sources, allowing for controlled clipping at desired voltage levels. These circuits play a crucial role in waveform modification, ensuring signal integrity and preventing voltage spikes in electronic systems.
- **Implemented Schematic :**

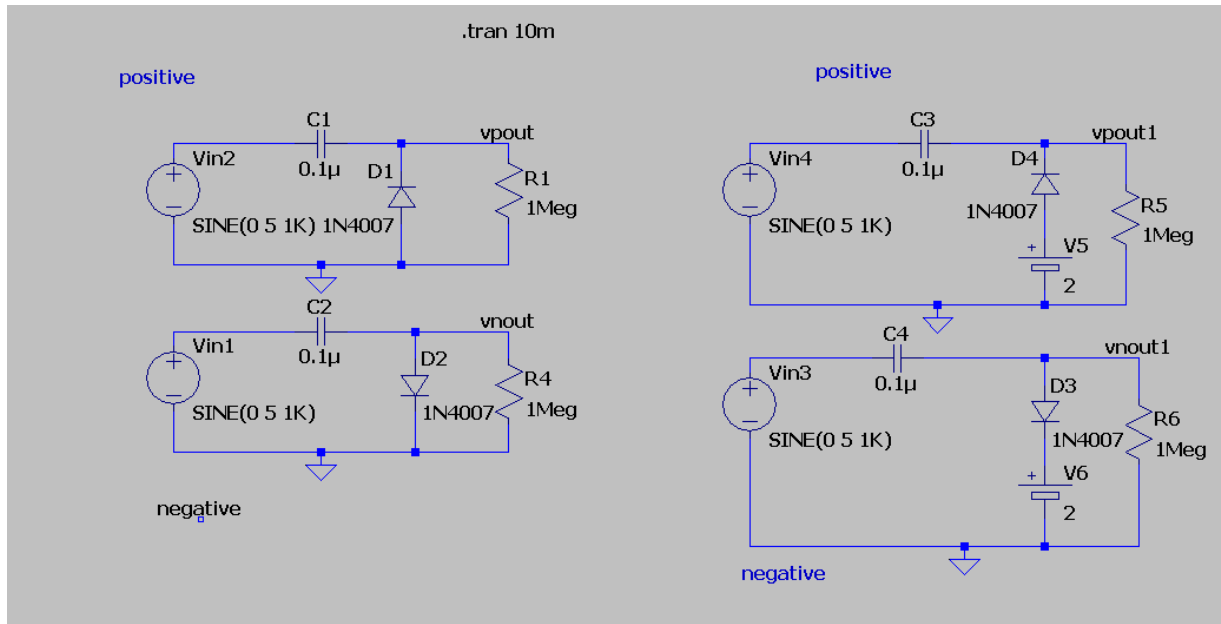


- Obtained Waveforms

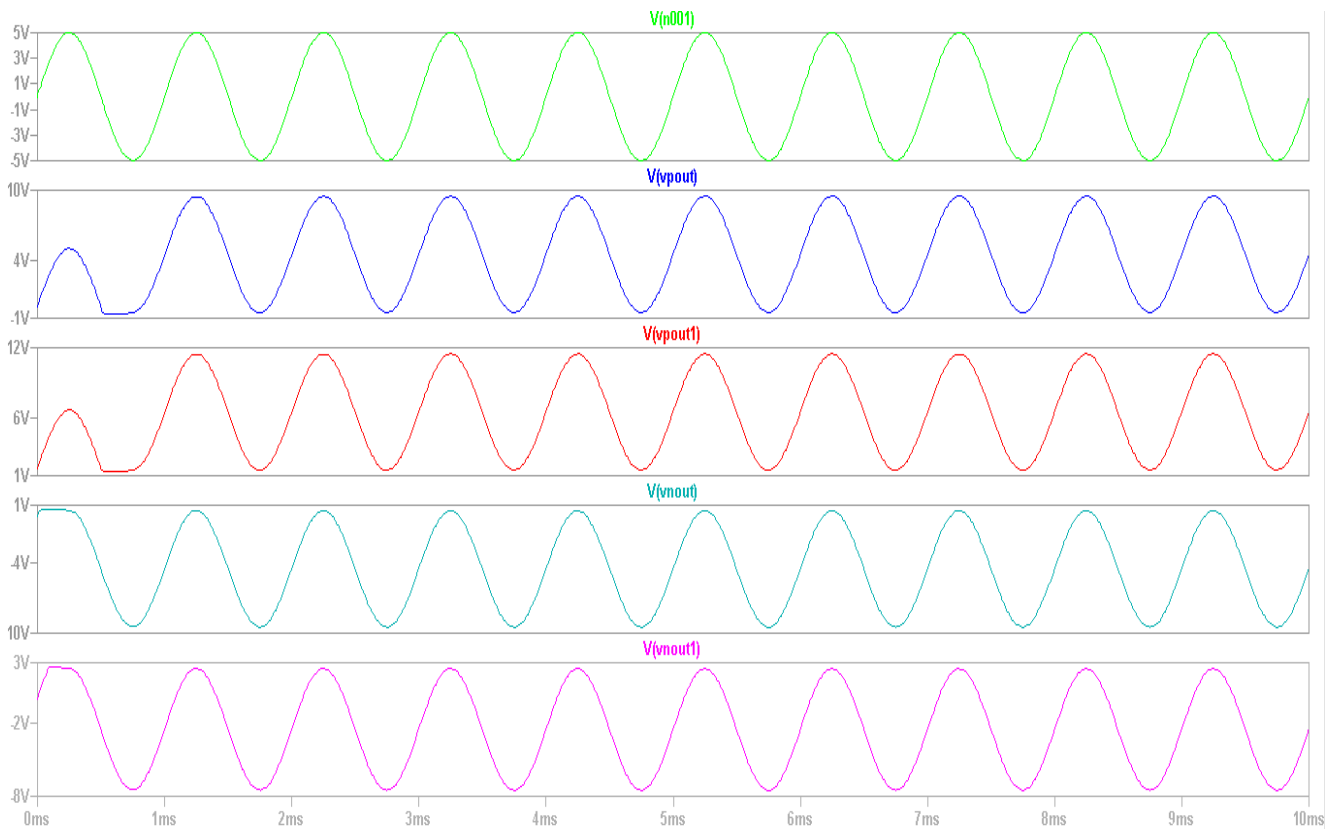


EXPERIMENT- 3

- **Aim :** To design a voltage clamper circuit using positive and negative voltage reference.
- **Software used :** Lt-spice
- **Theory :** A voltage clamper is a circuit that shifts the entire waveform to a different DC level without changing its shape. Clamping circuits are widely used in signal processing, television receivers, and communication systems to fix the DC level of signals. The two main types of clamper circuits are positive clammers and negative clammers, which depend on whether the circuit shifts the signal in the positive or negative direction.
 - **Positive Clamper :** A positive clamper shifts the waveform upward so that the lowest voltage point of the signal becomes zero or a positive value. It consists of a diode, a capacitor, and a resistor. The diode conducts during the negative half-cycle, allowing the capacitor to charge. During the positive half-cycle, the diode is reverse biased, and the stored charge in the capacitor shifts the waveform upward by the reference voltage. A positive clamper ensures that the output signal remains above a specific voltage level.
 - **Negative Clamper :** A negative clamper shifts the waveform downward, making the highest voltage point of the signal zero or a negative value. The circuit works similarly to the positive clamper but with reversed diode polarity. During the positive half-cycle, the diode conducts, charging the capacitor. In the negative half-cycle, the diode becomes reverse biased, and the capacitor shifts the waveform downward by the reference voltage.
 - By introducing a biasing voltage source, the clamping level can be adjusted to a desired reference voltage. These circuits are essential in applications where signals need to be adjusted to operate within specific voltage limits.
- **Implemented Schematic :**



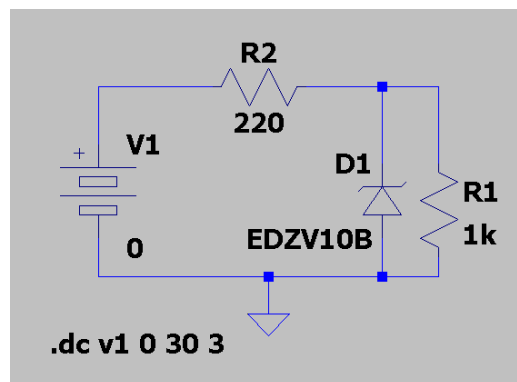
- Obtained waveforms :



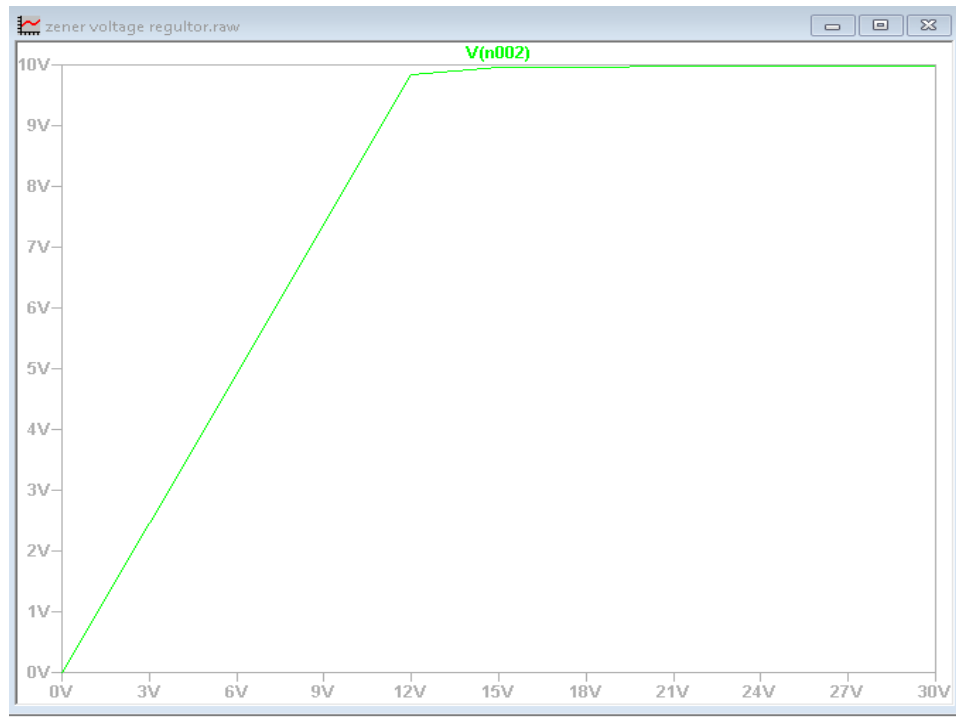
EXPERIMENT – 4

- Aim :** To design a voltage regulator circuit using a zener diode.

- **Software used :** Lt-spice
- **Theory :** A voltage regulator is a circuit designed to maintain a stable output voltage regardless of variations in input voltage or load conditions. A Zener diode is commonly used for voltage regulation due to its ability to operate in the reverse breakdown region, where it maintains a nearly constant voltage across its terminals.
- **Working Principle :** In a Zener voltage regulator circuit, the Zener diode is connected in reverse bias across the load, along with a series resistor. When the input voltage exceeds the Zener breakdown voltage, the diode conducts, maintaining a fixed voltage across the load. The series resistor limits the current flowing through the diode, preventing damage.
- **Types of Zener Voltage Regulators :**
 1. **Shunt Voltage Regulator:** The Zener diode is connected in parallel with the load, regulating the voltage by shunting excess current. It is simple and widely used in low-power applications.
 2. **Series Voltage Regulator:** Uses a transistor in series with the load, where the Zener diode provides a reference voltage to control the transistor's operation, offering better efficiency.
- **Applications :** Zener voltage regulators are used in power supplies, voltage reference circuits, and overvoltage protection to ensure that electronic components receive a stable voltage, preventing damage due to fluctuations.
- **Implemented Schematic :**

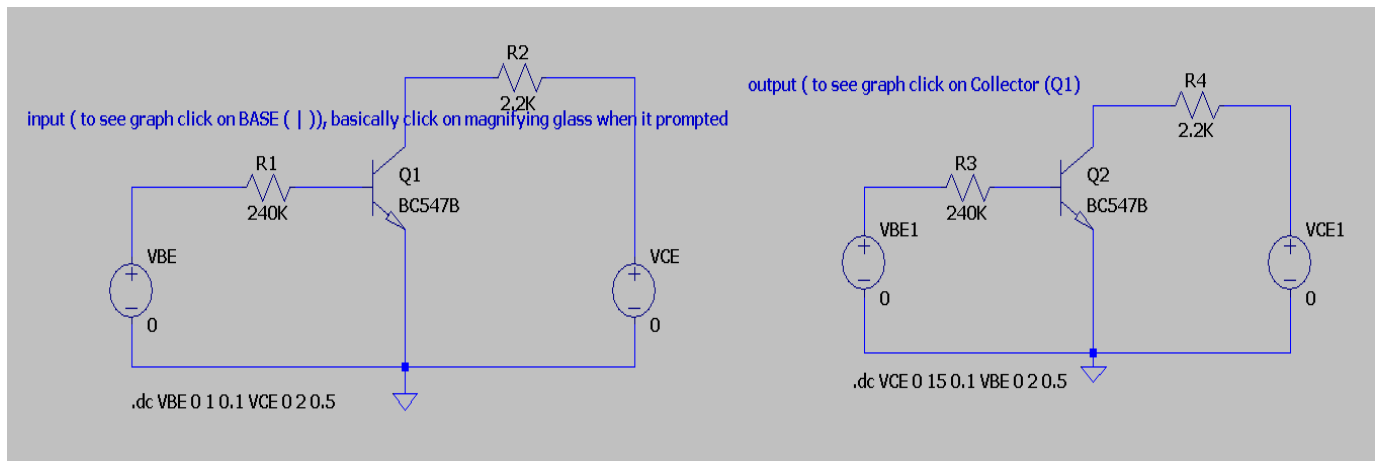


- **Obtained waveforms :**

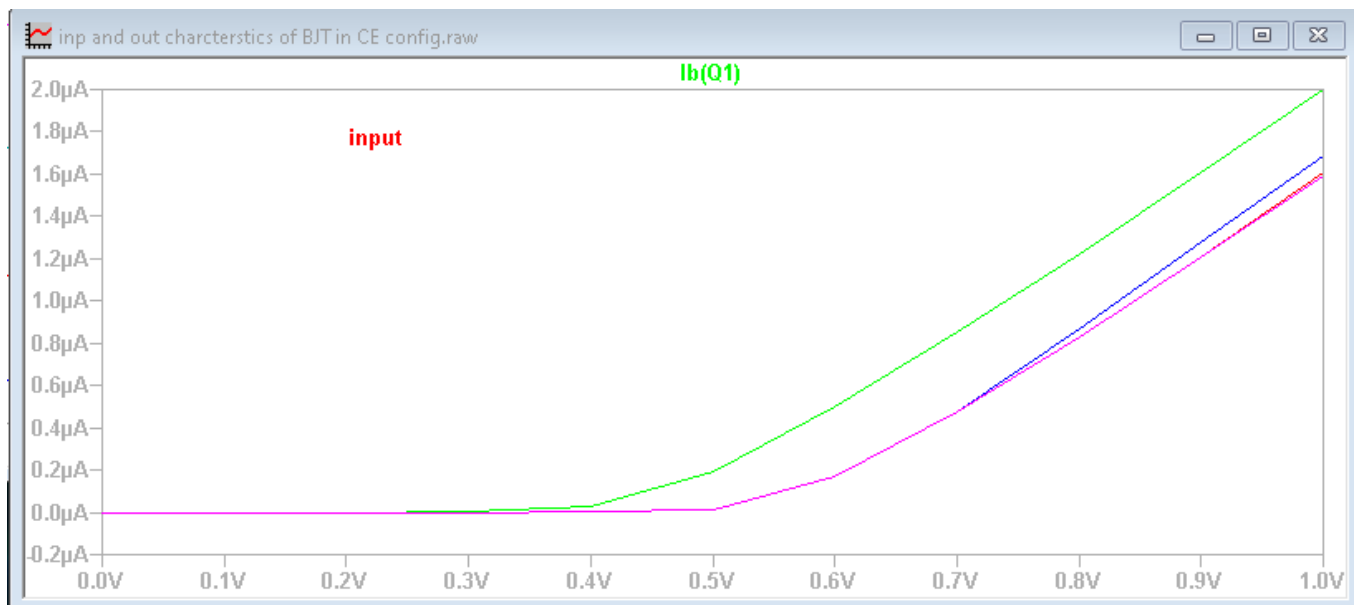


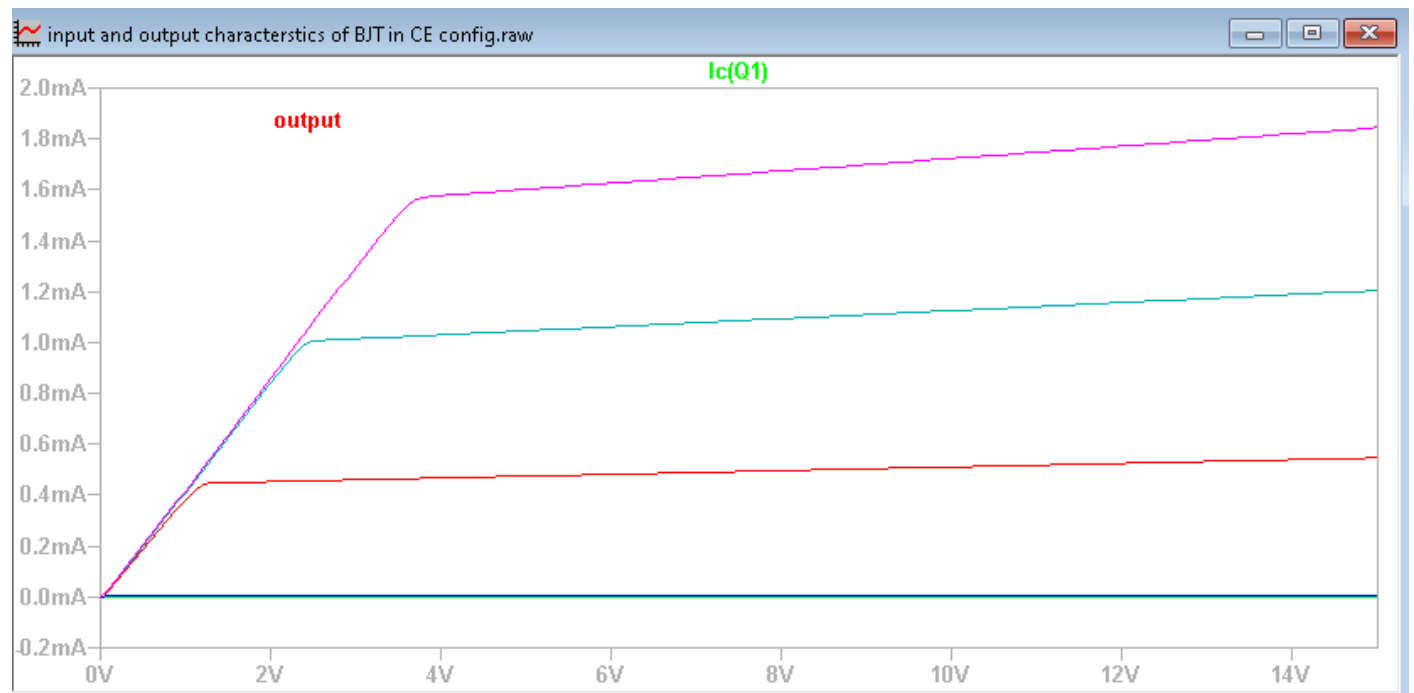
EXPERIMENT – 5

- **Aim :** To plot the input and output characteristics of BJT in CE configuration.
- **Software used :** Lt-spice
- **Theory :** A Bipolar Junction Transistor (BJT) is a three-terminal semiconductor device used for amplification and switching applications. The Common Emitter (CE) configuration is the most widely used transistor configuration due to its high gain and efficiency. In this configuration, the emitter terminal is common to both the input and output circuits.
 - **Input Characteristics :** The input characteristics describe the relationship between the base current (I_b) and the base-emitter voltage (V_{BE}), while keeping the collector-emitter voltage (V_{CE}) constant. It resembles the behavior of a forward-biased PN junction, where a small increase in V_{BE} results in an exponential increase in I_b . The input impedance of a CE configuration is relatively low.
 - **Output Characteristics :** The output characteristics show the variation of collector current (I_c) with collector-emitter voltage (V_{CE}), for different values of base current (I_b). These characteristics are divided into three regions:
 1. **Cutoff Region** – The transistor is OFF, and I_c is nearly zero.
 2. **Active Region** – The transistor operates normally, and I_c is controlled by I_b , making it suitable for amplification.
 3. **Saturation Region** – Both junctions are forward biased, and I_c reaches its maximum value.
 - **Applications :** The CE configuration is widely used in amplifiers, switches, and signal processing circuits due to its high current gain (β), good voltage gain, and moderate input impedance.
- **Implemented Schematic :**



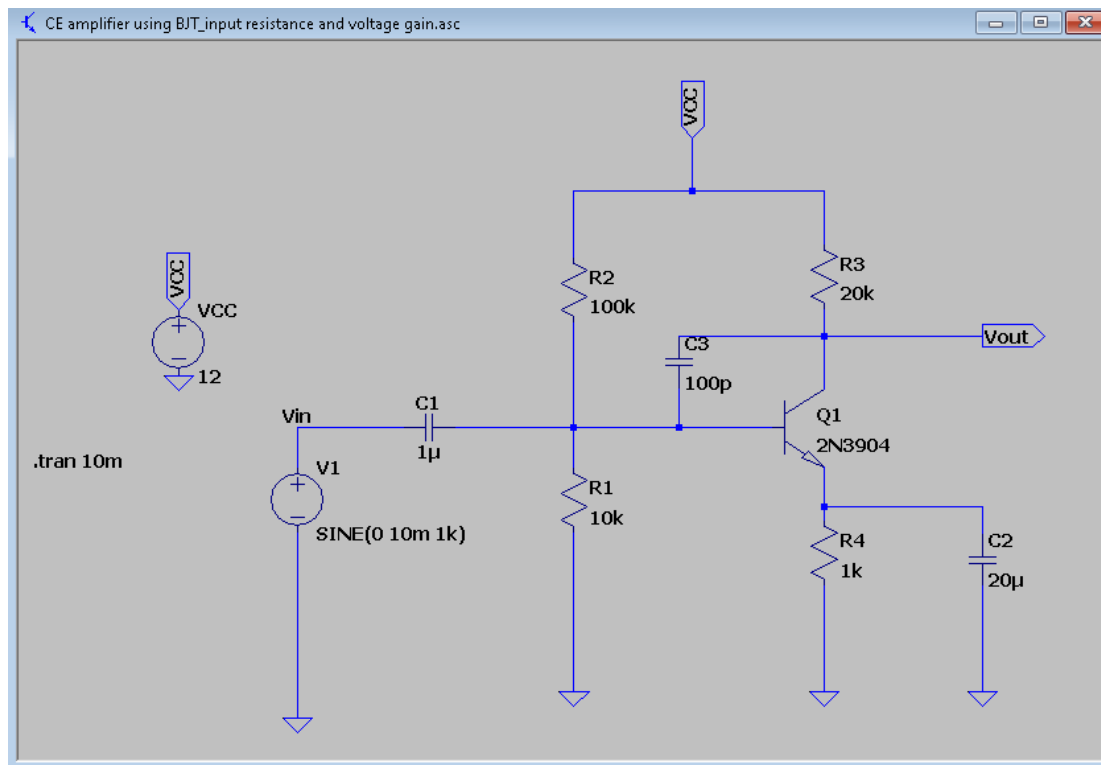
- **Obtained waveforms :**

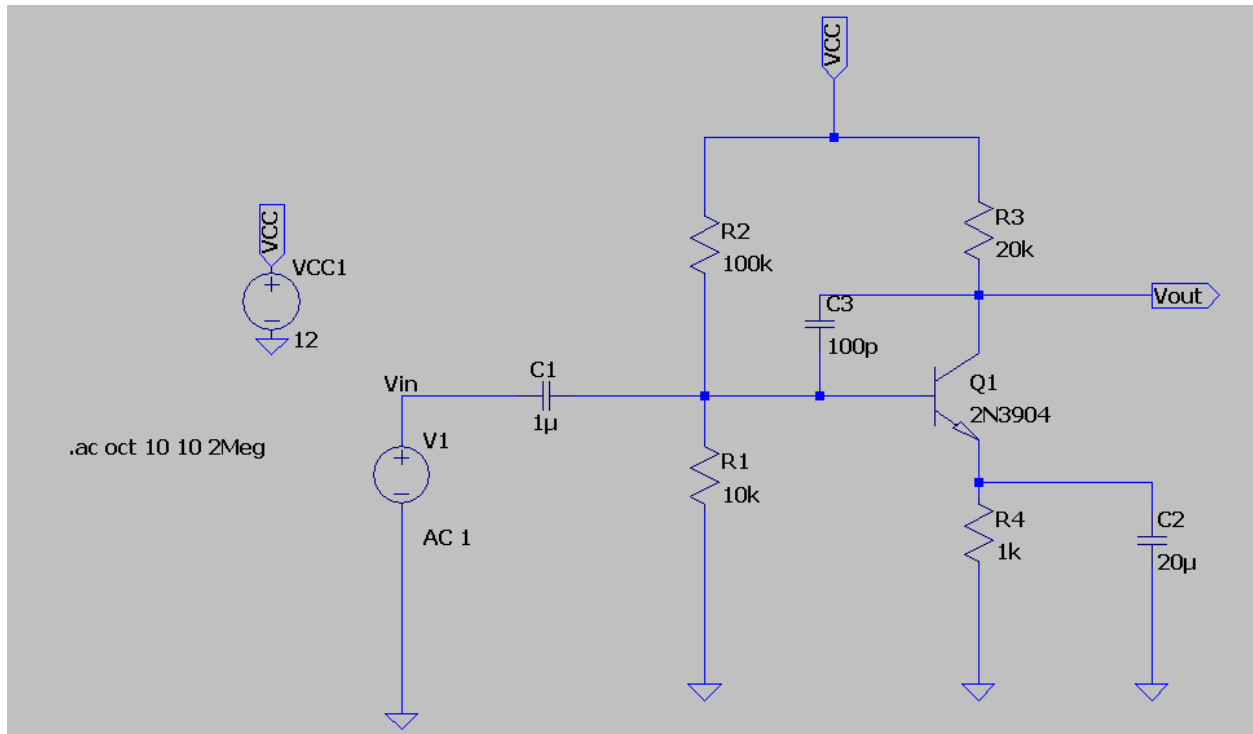




EXPERIMENT – 6

- **Aim :** Design CE amplifier using BJT: measure input resistance, voltage gain and plot the frequency response curve and measure bandwidth.
- **Software used :** Lt-spice
- **Theory :** The CE Amplifier provides high gain & wide frequency response. The emitter lead is common to both input & output circuits and is grounded. The emitter-base circuit is forward biased. The collector current is controlled by the base current rather than emitter current. The input signal is applied to base terminal of the transistor and amplifier output is taken across collector terminal. A very small change in base current produces a much larger change in collector current. When +ve half cycle is fed to the input circuit, it opposes the forward bias of the circuit which causes the collector current to decrease, it decreases the voltage more –ve. Thus when input cycle varies through a –ve half cycle, increases the forward bias of the circuit, which causes the collector current to increase thus the output signal is common emitter amplifier is in out of phase with the input signal.
- **Implemented Schematic :**





- Obtained Waveforms :

